

Day 115 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Geography of India (11): Himalayan Disaster-Preparedness Infrastructure Expansion

READING MATERIAL

Covers expanded Himalayan disaster-monitoring infrastructure, announced at the World Summit on Disaster Management (Dehradun, November 2025), and the glacial-lake hazard it responds to.

Q: What is a Glacial Lake Outburst Flood (GLOF), and why is its risk increasing?

A: A sudden, often catastrophic release of water from a glacial lake, threatening downstream communities. Risk is increasing because climate-change-driven glacier recession is forming new glacial lakes (meltwater pooling between the retreating glacier and its frontal moraine) and causing existing lakes to expand/merge.

Q: Where and when was the World Summit on Disaster Management held, where expanded Himalayan monitoring infrastructure was announced?

A: Dehradun, November 2025.

Q: What expanded monitoring infrastructure did the Union Minister of State for Earth Sciences announce at this summit?

A: 33 meteorological observatories, 142 automatic weather stations, 107 rain gauges, and 3 weather radars (with 3 more planned).

Q: What specialised programme and alert system were also announced/expanded?

A: A specialised Himalayan climate study programme was launched; and the 'Nowcast' alert system (providing a 3-hour forecast, already used in major metros) is being expanded across Uttarakhand.

MOCK TEST (WITH ANSWERS)

1. A Glacial Lake Outburst Flood (GLOF) refers to:

- A. A monsoon-season river flood
- B. A sudden, catastrophic release of water from a glacial lake ✓**
- C. Annual snowmelt in spring
- D. A man-made dam failure only

Explanation: A sudden, catastrophic release of water from a glacial lake. [medium, web-research]

2. The World Summit on Disaster Management, where expanded Himalayan monitoring infrastructure was announced, was held in:

- A. Shimla
- B. Dehradun ✓**
- C. Gangtok
- D. Srinagar

Explanation: Dehradun (November 2025). [medium, web-research]

3. How many automatic weather stations were announced as part of expanded Himalayan monitoring infrastructure?

- A. 42
- B. 82
- C. 142 ✓**
- D. 182

Explanation: 142. [hard, web-research]

4. The 'Nowcast' alert system, being expanded across Uttarakhand, provides a forecast for how many hours ahead?

- A. 1 hour
- B. 3 hours ✓**
- C. 6 hours
- D. 24 hours

Explanation: 3 hours. [medium, web-research]

5. GLOF risk is increasing primarily due to:

A. Increased rainfall only

B. Climate-change-driven glacier recession forming/expanding glacial lakes ✓

C. Deforestation

D. Increased tourism

Explanation: Climate-change-driven glacier recession. [medium, web-research]

Part B – Aptitude & Mental Ability: Highest Common Factor (HCF)

READING MATERIAL

HCF is 1 of 10 named Aptitude sub-skills. Real exam evidence (4 confirmed hits across 2022/2024/2025) shows TNPSC favors HCF *application* problems — measuring/tiling/container word problems and the HCF-LCM relationship — over plain 'find the HCF of these two numbers' questions, so the reading below leads with those application shapes.

Q: Real exam (2025): Two numbers are in the ratio 15:11. If their HCF is 13, find the numbers.

A: When two numbers are in a given ratio, write them as $15k$ and $11k$. Since $\text{HCF}(15,11)=1$ (the ratio is already in lowest terms), k IS the HCF — so $k=13$. The numbers are $15 \times 13=195$ and $11 \times 13=143$.

Q: Real exam (2025): What is the least number of square tiles needed to pave the ceiling of a room 15m 17cm long and 9m 2cm broad?

A: Convert to one unit: $1517 \text{ cm} \times 902 \text{ cm}$. The LARGEST square tile that exactly fits (no cutting) has a side equal to the HCF of the two dimensions. $\text{HCF}(1517, 902) = 41 \text{ cm}$ (via the Euclidean algorithm: $1517=1 \times 902+615$; $902=1 \times 615+287$; $615=2 \times 287+41$; $287=7 \times 41+0$). Tile area = $41 \times 41 = 1,681 \text{ cm}^2$. Room area = $1517 \times 902 = 1,368,334 \text{ cm}^2$. Number of tiles = $1,368,334 \div 1,681 = 814$.

Q: Real exam (2024): If the HCF of x^3-8y^3 and $x^2-k^2y^2$ is $x+ky$, find k .

A: Factor x^3-8y^3 using $a^3-b^3=(a-b)(a^2+ab+b^2)$ with $a=x$, $b=2y$: $x^3-8y^3 = (x-2y)(x^2+2xy+4y^2)$. Factor $x^2-k^2y^2$ using $a^2-b^2=(a-b)(a+b)$: $x^2-k^2y^2 = (x-ky)(x+ky)$. For ' $x+ky$ ' to be the common factor (HCF) of both, it must match one of the first expression's factors. $(x-2y)$ is the only linear factor available there, so $x+ky$ must equal $x-2y$, giving $k=-2$.

Q: Real exam (2022): What is the greatest possible volume of a vessel that can exactly measure the milk in cans of 80, 100, and 120 litres (each can measured to full capacity only)?

A: This is asking for the HCF of 80, 100, and 120. $\text{HCF}(80,100)=20$, then $\text{HCF}(20,120)=20$. So the answer is 20 litres — the largest vessel that can measure each can's full volume an exact whole number of times.

Q: Real exam (2022): The HCF and LCM of two numbers are 16 and 240 respectively. If one number is 48, find the other.

A: Use the relationship: $\text{HCF} \times \text{LCM} = \text{product of the two numbers}$. $16 \times 240 = 3,840$. Other number = $3,840 \div 48 = 80$.

Q: Method — finding HCF via the Euclidean algorithm (the fast way for large numbers): What is the HCF of 1517 and 902?

A: Repeatedly divide the larger by the smaller and replace the larger with the remainder, until the remainder is 0 — the last nonzero remainder is the HCF. $1517=1 \times 902+615 \rightarrow 902=1 \times 615+287 \rightarrow 615=2 \times 287+41 \rightarrow 287=7 \times 41+0$. $\text{HCF}=41$. This is the same calculation used in ro2 above — recognize it as the standard fast method rather than listing all factors of both numbers.

MOCK TEST (WITH ANSWERS)

1. Two numbers are in the ratio 21:14. If their HCF is 12, find the larger number.

A. 36 ✓

B. 24

C. 252

D. 168

Explanation: Watch out: 21 and 14 are NOT coprime ($\text{gcd}=7$), so you can't directly multiply them by the HCF. Reduce the ratio first: $21:14 = 3:2$ (coprime). Now apply the HCF: larger = $3 \times 12=36$, smaller = $2 \times 12=24$. [hard, authored]

2. Find the largest square tile that can exactly pave a floor measuring 3m 8cm by 4m 20cm (no cutting).

A. 28 cm ✓

B. 14 cm

C. 56 cm

D. 84 cm

Explanation: Convert to cm: 308 and 420. $\text{HCF}(308,420)$: $420=1 \times 308+112$; $308=2 \times 112+84$; $112=1 \times 84+28$; $84=3 \times 28+0$. $\text{HCF}=28 \text{ cm}$. [hard, authored]

3. If the HCF of a^2-4b^2 and $a^2+ab-6b^2$ is $a-2b$, this is consistent with which factorization of the second expression?

A. $(a+3b)(a-2b)$ ✓

B. $(a-3b)(a+2b)$

C. $(a+3b)(a+2b)$

D. $(a-3b)(a-2b)$

Explanation: $a^2-4b^2=(a-2b)(a+2b)$. The second expression factors as $(a+3b)(a-2b) = a^2+ab-6b^2$ — sharing the common factor $(a-2b)$, so that's the HCF. [hard, authored]

4. What is the greatest volume that can exactly measure the contents of drums holding 72, 96, and 120 litres?

A. 12 litres

B. 24 litres ✓

C. 8 litres

D. 6 litres

Explanation: $\text{HCF}(72, 96) = 24$, then $\text{HCF}(24, 120) = 24$. Answer: 24 litres. [medium, authored]

5. The HCF and LCM of two numbers are 12 and 180 respectively. If one number is 36, find the other.

A. 48

B. 54

C. 60 ✓

D. 72

Explanation: $\text{HCF} \times \text{LCM} = 12 \times 180 = 2,160$. Other number = $2,160 \div 36 = 60$. [medium, authored]

6. Using the Euclidean algorithm, find the HCF of 1,073 and 925.

A. 37 ✓

B. 29

C. 31

D. 43

Explanation: $1073 = 1 \times 925 + 148$; $925 = 6 \times 148 + 37$; $148 = 4 \times 37 + 0$. $\text{HCF} = 37$. [medium, authored]

7. Two numbers are in the ratio 8:9 and their HCF is 7. Find their sum.

A. 105

B. 119 ✓

C. 126

D. 112

Explanation: Numbers: $8 \times 7 = 56$ and $9 \times 7 = 63$. Sum = $56 + 63 = 119$. [medium, authored]

Part C – General English: Prose: "The Attic" — Satyajit Ray

READING MATERIAL

A first-person short story by Satyajit Ray (better known as a filmmaker) about Aditya, the narrator's friend and business partner, who revisits his ancestral hometown of Brahmapur after many years and finally confesses to a childhood theft — taking a medal belonging to a boy named Sasanka Sanyal — that has weighed on his conscience for 29 years.

Q: Who wrote "The Attic", and what is the central theme of the story?

A: Satyajit Ray. Central theme: human flaws, guilt, and the power of confession/forgiveness — Aditya finally confesses to a childhood theft that has burdened his conscience for decades.

Q: What was Aditya's relationship to the narrator, and what was his ancestral hometown?

A: Aditya was the narrator's friend and business partner; together they had newly started a factory in Deodarganj. Aditya's ancestral hometown (which they revisit during the story) was Brahmapur.

Q: What had Aditya done 29 years earlier that he finally confesses to, and who was the victim?

A: As a child, Aditya had stolen a medal — a 'Special Prize for Recitation' (engraved 'Sriman Sasanka Sanyal — Special Prize for Recitation — 1948') — belonging to a boy named Sasanka Sanyal, and had hidden it in the attic of his ancestral house. He confesses this theft to Sanyal after 29 years, and the medal is finally restored to its rightful owner.

Q: What was special about the attic, and what objects from childhood were found there?

A: The attic was Aditya's favourite childhood place, a private world of his own (a common motif — attics being favoured spaces for children's imagination). Among the objects found there were a broken cricket bat, the remains of an armchair, and a wooden packing case.

Q: What other factual details about the setting does the story establish — the house's age, the season of the visit, and a notable nearby landmark?

A: Aditya's ancestral house was about 200 years old. The visit took place in the Bengali month of Maha (January/February). Opposite the tea shop in the town (owned by 'Nagen Uncle') stood a Shiva temple.

Q: What was Aditya's father's occupation?

A: He was a landlord.

MOCK TEST (WITH ANSWERS)

1. "The Attic" was written by:

- A. R.K. Narayan
- B. Rabindranath Tagore
- C. Satyajit Ray ✓**
- D. Mulk Raj Anand

Explanation: Satyajit Ray. [easy, web-research]

2. Aditya's ancestral hometown, which he revisits in the story, was:

- A. Brahmapur ✓**
- B. Deodarganj
- C. Kolkata
- D. Delhi

Explanation: Brahmapur. [medium, web-research]

3. What had Aditya done 29 years earlier that he finally confesses to in the story?

- A. Fought with a friend
- B. Stolen a book
- C. Stolen someone's medal ✓**
- D. Run away from school

Explanation: He had stolen Sasanka Sanyal's recitation medal. [medium, web-research]

4. Who was the rightful owner of the medal that Aditya had stolen and hidden in the attic?

- A. Nagen Uncle

B. Sasanka Sanyal ✓

C. Jogesh Kabiraj

D. Aditya's own brother

Explanation: Sasanka Sanyal. [medium, web-research]

5. Aditya's ancestral house, where the attic was located, was approximately how old?

A. 50 years

B. 100 years

C. 200 years ✓

D. 300 years

Explanation: About 200 years. [medium, web-research]

6. During which season/month did Aditya and the narrator's visit to Brahmapur take place?

A. Summer

B. Maha (January/February) ✓

C. Monsoon

D. Autumn

Explanation: Maha (Jan/Feb). [hard, web-research]

7. What stood opposite the tea shop in the town?

A. A school

B. A bank

C. A hospital

D. A Shiva temple ✓

Explanation: A Shiva temple. [medium, web-research]

8. The tea shop in the town was owned by:

A. Sanyal

B. Kabiraj

C. Nagen Uncle ✓

D. Aditya's uncle

Explanation: Nagen Uncle. [medium, web-research]

9. What was Aditya's favourite childhood place in his ancestral house?

A. The library

B. The attic ✓

C. The garden

D. The kitchen

Explanation: The attic. [easy, web-research]

10. Aditya's father's occupation was:

A. Teacher

B. Landlord ✓

C. Police officer

D. Priest

Explanation: Landlord. [medium, web-research]